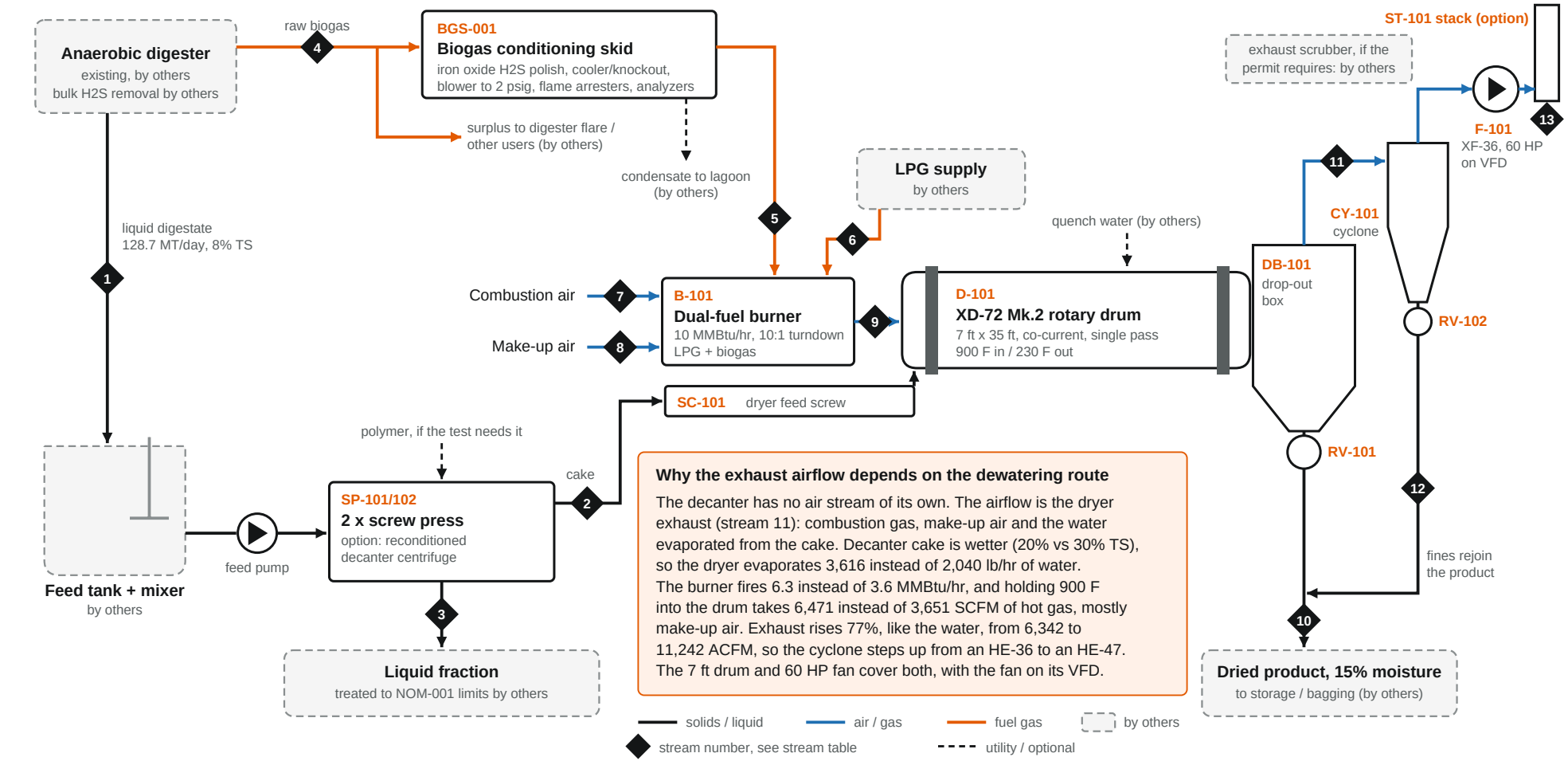
PROCESS FLOW DIAGRAM

LETEK DCG - CHICKEN MANURE DIGESTATE DRYING - 128.7 MT/DAY

MCE-LETEK-PFD-001 Rev C - Oct 19, 2026 - for budgetary proposal 20261019-LETEK-C



KEY NUMBERS BY DEWATERING ROUTE (DRYER AT FULL SOLIDS CAPTURE)

	Cake to dryer	Water evaporated	Dryer duty	Biogas available vs duty	Exhaust (stream 11)	Fan shaft power	Cyclone	Liquid fraction (typical capture)
Screw press (base)	3,153 lb/hr, 30% TS	2,040 lb/hr	3.57 MMBtu/hr	about 170%	**6,342 ACFM**	about 18 bhp	HE-36	101 to 111 MT/day
Decanter (option)	4,729 lb/hr, 20% TS	3,616 lb/hr	6.33 MMBtu/hr	about 96%	**11,242 ACFM**	about 33 bhp	HE-47	82 to 93 MT/day

STREAM TABLE AND EQUIPMENT

SCREW PRESS BASE AND DECANTER OPTION

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STREAMS

#	Stream	Screw press, 30% TS cake (base)	Decanter, 20% TS cake (option)
1	Liquid digestate to dewatering	11,822 lb/hr (128.7 MT/day, about 34,000 gal/day), 8% TS, 946 lb/hr solids	same
2	Dewatered cake to dryer (design: full solids capture)	3,153 lb/hr at 30% TS (2,207 lb/hr water)	4,729 lb/hr at 20% TS (3,783 lb/hr water)
3	Liquid fraction to treatment (by others)	101 to 111 MT/day at typical capture (60 to 80%, no polymer, 30 to 35% TS cake), 2.0 to 3.8% TS. 8,669 lb/hr (94.4 MT/day) at full capture	82 to 93 MT/day at typical capture (70 to 90%), 1.2 to 3.3% TS. 7,093 lb/hr (77.2 MT/day) at full capture
4	Raw biogas from digester	195 scfm (7,455 Nm3/day), 55% CH4, 500 ppm H2S or less; 6.07 MMBtu/hr average	same
5	Conditioned biogas to burner (H2S under 100 ppm, 2 psig)	about 115 scfm (3.57 MMBtu/hr); the other 80 scfm goes to the digester flare / other users ahead of the skid	all 195 scfm (6.07 MMBtu/hr, covers about 96% of the duty)
6	LPG to burner (start-up, make-up, digester down)	none at average biogas; 179 lb/hr on LPG alone	about 13 lb/hr make-up at average biogas (more if the biogas heating value is lower; analysis pending); 318 lb/hr on LPG alone
7	Combustion air, 85 F	821 ACFM	1,456 ACFM
8	Make-up (dilution) air, 85 F	2,797 SCFM	4,958 SCFM
9	Mixed gas into drum, 900 F	3,651 SCFM	6,471 SCFM
10	Dried product, 180 F	1,113 lb/hr at 15% moisture (946 solids + 167 water); 12.1 MT/day at full capture	same
11	Dryer exhaust to cyclone, 230 F	**6,342 ACFM**, carrying 2,040 lb/hr evaporated water	**11,242 ACFM**, carrying 3,616 lb/hr evaporated water
12	Cyclone fines (rejoin product)	included in stream 10	included in stream 10
13	Exhaust to stack	6,342 ACFM at about 230 F; 12 in WC fan static, about 18 bhp	11,242 ACFM; about 33 bhp
	Dryer duty (design / rigorous check)	3.57 / 3.24 MMBtu/hr	6.33 / 5.70 MMBtu/hr
	Cyclone (MCE cyclone calculator)	HE-36 budget pick, range 6,175 to 7,600 CFM; final pick after the sample test	HE-47 budget pick, range 10,825 to 12,600 CFM; final pick after the sample test

EQUIPMENT

	Feed tank and mixer	By others. Good mixing ahead of dewatering is essential; mixer sized after the sample test.
SP-101/102	**Dewatering, base**	2 screw presses in parallel with the chopper feed pump, VFD drives and basic controls, subject to a sample test. Option: reconditioned decanter centrifuge.
SC-101	**Dryer feed screw**	2 HP. The dewatering unit discharges by chute onto the feed screw (no storage hopper).
B-101	**Dual-fuel burner**	10 MMBtu/hr, LPG + biogas, two NFPA 86 trains, automatic changeover to LPG on loss of biogas.
D-101	**XD-72 Mk.2 rotary drum**	7 ft x 35 ft, co-current single pass, A36 (304 SS option recommended), 20 HP drive. Design gas capacity about 13,470 ACFM.
DB-101 / RV-101	**Drop-out box and product airlock**	Heavy duty drop-out box; airlock 1.5 HP.
CY-101 / RV-102	**Cyclone and fines airlock**	HE-36 in 304 SS (HE-47 with the decanter); FT-12 airlock. Fines rejoin the product.
F-101	**Exhaust fan and VFD**	XF-36, 60 HP inverter-duty, 12 in WC, on a 60 HP VFD trimmed on drum outlet draft, so the exhaust airflow follows the drying load; ducting up to 60 ft.
ST-101	**Exhaust stack (option)**	24 in OD 304 SS, 10 ga, discharge about 35 ft above grade, EPA Method 1 sampling ports, guyed. Exhaust scrubber, if the air permit requires one, by others.
BGS-001	**Biogas conditioning skid**	For 195 scfm at 500 ppm H2S or less: iron oxide vessels, gas cooler and knockout, blower, flame arresters, analyzers, control panel.
	Automation	PLC / HMI with LPG and biogas fuel profiles, dryer temperature control, links to dewatering, skid and plant.

Basis: 128.7 MT/day of liquid digestate at 8% TS, 24 h/day; grit and sand handled separately. Dryer streams are from the MCE Dryer HMB Calculator (MCE-HMB-2026-002 Rev C) at full solids capture, the largest dryer load; air and LPG figures are on its propane basis. At typical press capture (60 to 80%) and 30 to 35% TS cake the exhaust is about 3,000 to 5,100 ACFM; the cyclone and fan control are confirmed after the sample test, which also sets cake solids, solids capture and polymer demand. Liquid fraction composition is given in the proposal.